KNOWN DEVELOPMENT CASE

Log Insights

Control-stream failures during BGL job startup

Engineer diagnosis / Team investigation / Manager allocation

Analyst-written walkthrough from real tool calls.
No new Claude answer or quality-evaluation result.

APPROACH AND SYSTEM

Shared tools and evidence

CLIENTS

Claude Code / Desktop
MCP over stdio

Web chat
AI SDK adapter

↔

SHARED REGISTRY

Zod contracts
Parameterized SQL
Limits and fact IDs

↔

SERENEDB

BGL + OpenStack
Separate systems

Search, counts, trends
episodes and percentiles

RETRIEVE

search_logs / get_log_context

Original bounded evidence.

AGGREGATE

count_events / recurring_patterns

Counts over the stated population.

VERIFY

check_claims

Submitted values, metrics and groups.

The database computes analytical values. Claude explains in the product. This capture invokes the same registry directly, without a model call.

QUERY EVIDENCE

Full populations, bounded original examples

BGL / FULL LOADED SPAN

4,747,945 r_8f3bdd12.f4 log events

OPENSTACK / FULL LOADED SPAN

207,820 r_2c4302ad.f4 log events

APPREAD / REQUESTED BGL WINDOW

5,983 r_bcd0408f.f3 events
7 r_bcd0408f.f8 episodes

ciod: failed to read message prefix on control stream (CioStream socket to 172.16.96.116:33399

Original event 14738 / search is truncated.

INDEPENDENT OPENSTACK EXAMPLE

2 r_872d82bd.f8 5xx accesses
out of 98,733 r_872d82bd.f6 API accesses

Bad response code while validating token: 503

Original event 10070743 / complete same-request context.

APPREAD scope: 2005-06-01T00:00:00Z r_bcd0408f.f15 to 2005-07-01T00:00:00Z r_bcd0408f.f16, end exclusive, UTC. OpenStack uses its own full span. No shared time axis.

ROLE DECISIONS

Investigation changes with the role

ENGINEER

Locate the origin

512 r_bcd0408f.f12 emitting nodes. Broad symptoms leave the initiating component open.

NEXT CHECK

Match job-start inputs and filesystem availability.

TEAM LEAD

Investigate recurrence

Program-image errors: 22 r_8207d0e9.f16 episodes. Previous-period trend is unavailable.

TEAM CHECK

Workload Runtime reviews startup validation.

MANAGER

Allocate effort

Team ranking favors program-image errors by episodes and breadth.

DECISION

Prioritize the startup check over line volume alone.

Same BGL window. Ownership is fictional. The team ranking is truncated but evaluates all candidates. Business impact unknown. These are analyst-written views, not client acceptance.

EXECUTED DISCRIMINATING CHECK

Inspect job-start inputs before network repair

ORIGINAL ASSESSMENT

A systemic job/network event at the I/O nodes was suspected.

Recorded before the original full-corpus profiling. Untested background.

EXECUTED PRECURSOR CHECK

Missing-directory errors precede
6 r_16d9e58b.f6 of
7 r_16d9e58b.f1 episode-start anchors.

June-scoped background rate:
0.00590349 r_16d9e58b.f7.

CHANGED NEXT STEP

Check the working directory and program image at startup.

Temporal association: supported.
Initiating cause: open.
No repair was performed.

The context preserves an earlier chdir error on another node. Broad symptoms and temporal precursors do not prove a shared job, a network defect or causality.

What remains unknown

LIMITS OF THIS CONCLUSION

Initiating node, job boundaries and why the working directory was unavailable remain open.

No downtime, cost or customer effect is established. Episodes are not incidents.

Formal baseline and holdout evaluation are deferred. This known case makes no model-quality comparison.

REVIEWABLE ARTIFACTS

English investigation

Inputs, original receipts and hashes

Checked values and groups

Repository visibility and signed-out access still need verification.

Known development/demo material. No claim of holdout performance, proven cause or complete role compliance. Original public log excerpts retain their Loghub attribution.

REPRODUCIBLE QUERY

Original SQL and a matching registry re-run

EXACT EXECUTED PHRASE-COUNT QUERY

```
SELECT count(*) AS total FROM log_events_idx WHERE
log_events_idx.message @@ ts_phrase($1) AND log_events_idx.dataset
= $2 AND log_events_idx.ts >= $3::TIMESTAMPTZ AND
log_events_idx.ts < $4::TIMESTAMPTZ
```

```
Bound parameters: (
  "failed to read message prefix",
  "bgl",
  "2005-06-01T00:00:00Z",
  "2005-07-01T00:00:00Z",
)
```

TOOL AND RESULT

search_logs

dataset: bgl / match: phrase

5,983 r_b63b6c6e.f4 exact matches

Bounded hits are returned separately.

Registry re-runs for the precursor, team-priority and OpenStack latency calls match signature, data version and result digest.

Exact SQL and parameters are read from the original receipt at build time. Data version: dv_229e422f86a9fc6e. Re-run invokes the registered tool with stored input, never stored SQL.

TOOL CATALOG

The shared registry

FIND AND INSPECT

describe_corpus
Population, fields, versions and capabilities

search_logs
Full-text query and bounded original hits

get_log_context
Event neighbours or precursor evidence

check_claims
Values, metrics and groups against fact IDs

MEASURE AND COMPARE

count_events
Grouped counts, shares and period comparison

event_trend
Buckets within a dataset span

recurring_patterns
Episodes, breadth and technical investigation order

latency_stats
OpenStack API-access percentiles and rates

SereneDB computes analytical values. TypeScript validates contracts, assigns fact IDs, stores receipts and checks claims. The core tools do not change the corpus.

VERIFICATION AND LIMITS

Verified facts, bounded conclusions

OBSERVED IN THIS CAPTURE

All cited values were submitted to the shared claim checker, including their metric and group.

The saved batches report no mismatch or unresolved fact. This does not check explanations or prove causality.

The manifest retains successful registry re-run comparisons.

DEFERRED OPTIONAL EVALUATION

Baseline versus playbooks, a first holdout and independent quality ratings have not run. No improvement in model quality is claimed.

End-to-end model latency, tokens and API costs were not measured here.

Historical database timings are separate receipt measurements, not a controlled model benchmark.

See submission/worked-investigation.md and submission/performance/. Archived query evidence does not establish autonomous answer quality or complete client compliance.